

Named Entity Recognition with GLiNER and Relation Extraction with LLMs

Notebook for the BioASQ Task GutBrainIE on Gut-Brain Interplay Information Extraction at CLEF 2025

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Abstract

Biomedical Information Extraction from Natural Language Processing (NLP) is one of the greatest challenges driving innovation in the biomedical scientific field. In this work, we present our implementation pipeline for the GutBrain shared task covering both Named Entity Recognition and Relation Extraction. For subtask 6.1 (NER), we fine-tuned the GLiNER framework on expert-annotated GutBrain datasets, achieving robust entity recognition between the predefined categories. For the RE subtasks (6.2.1-6.2.3), we injected entity markers into text and employed fine-tuned BiomedBERT and pubmed-bert classifiers to predict relations between entities. By exploring Precision-oriented, Recall-oriented, and balanced configurations, we identified the best setups for maximizing Precision, Recall, and F1 for each task. Finally, we show our results with scatter plots and discuss the trade-off each run offers.

Keywords

Information Extraction, Named Entity Recognition, GLiNER, Relation Extraction, Large Language Models

1. Introduction

In the biomedical field, large volumes of textual data are generated daily, such as electronic health records and biomedical literature. Extracting and structuring this information in an efficient way is crucial for improving healthcare quality, supporting clinical decision-making, and advancing medical research [1]. Natural Language Processing (NLP) techniques have therefore become fundamental tools in medical text mining and Information Extraction (IE).

A key subtask of information extraction is Named Entity Recognition (NER), which involves identifying and categorizing spans of text into predefined categories such as diseases, treatments, and anatomical entities [2]. Figure 1 illustrates an example of NER annotation in biomedical text.

Based on NER, Relation Extraction (RE) identifies and extracts relationships between named entities from the underlying content [3]. RE is crucial for facilitating the extraction of information from large datasets, particularly when the data is unstructured. It supports many downstream applications, such as transforming unstructured corpora into knowledge graphs, question answering, and automated document processing [4, 5, 6, 7].

The CLEF 2025 conference is the 16th edition of the Conference and Labs of the Evaluation Forum (CLEF), continuing the popular CLEF campaigns that have been running since 2000 and contributing to the systematic evaluation of information access systems through experimentation with shared tasks¹. The aims of the conference include investigating new evaluation methodologies, tackling emerging application domains, and stimulating community building and knowledge transfer. In particular, BioASQ 2025 Lab Task 6, known as GutBrainIE, promotes the development of Information Extraction systems by extracting named entities and the relations between them².

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¹<https://clef2025.clef-initiative.eu/>

²<https://hereditary.dei.unipd.it/challenges/gutbrainie/2025/>

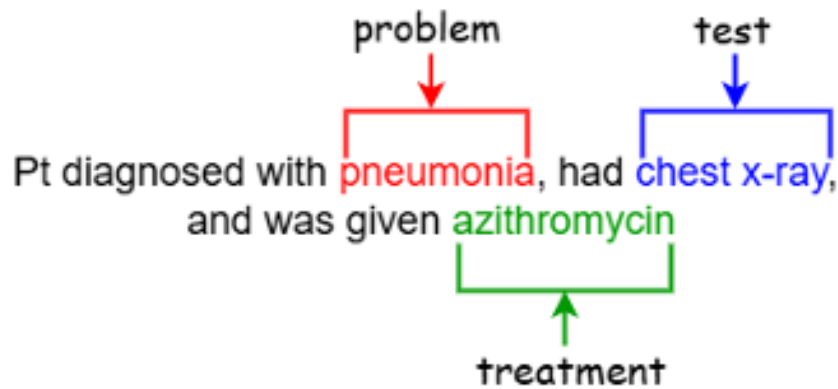


Figure 1: Example of Named Entity Recognition in a biomedical sentence. From [2].

Table 1

Example of Named Entity Recognition.

Label	Mention
Bacteria	alloprevotella
Chemical	short-chain fatty acids
DDF	irritable bowel syndrome

Table 2

Example of Binary Relation Extraction.

Subject Label	Object Label
Bacteria	Anatomical Location
Microbiome	DDF
Dietary Supplement	Microbiome

The challenge is divided into 4 subtasks:

- Task 6.1, the participants are provided with PubMed abstracts about gut-brain interplay and are asked to classify specific entity mentions into one of the 13 predefined categories: anatomical location, animal, biomedical technique, bacteria, chemical, dietary supplement, disease disorder or finding (DDF), drug, food, gene, human, microbiome and statistical technique. An example is shown in Table 1.
- The next challenges concerns Relation Extraction (RE) over the entities identified by the NER systems:
 - Task 6.2.1 is the Binary Relation Extraction, where participants are asked to identify which entities are in relation within a document. An example is shown in Table 2.
 - Task 6.2.2 is the Ternary Tag-Based Relation Extraction: participants are required to identify the actual entities involved in a relation and predict the type of relation. An example is shown in Table 3.
 - Finally, Task 6.2.3, the Ternary Mention-Based Relation Extraction, where participants are required to identify the actual entities involved in a relation and predict the type of relation. An example is shown in Table 4.

Table 3

Example of Ternary Tag-Based Relation Extraction.

Subject Label	Predicate	Object Label
Bacteria	Located In	Anatomical Location
Microbiome	Is linked to	DDF
Dietary Supplement	Influence	Microbiome

Table 4

Example of Ternary Tag-Based Relation Extraction.

Subject Text Span	Subject Label	Predicate	Object Text Span	Object Label
Microorganisms	Bacteria	Located in	Human Body	Anatomical Location
Microbiome	Microbiome	Is linked to	Psychiatric Disorders	DDF
Metformin	Dietary Supplement	Influence	DDF	Human Headache

2. Related Work

Early approaches to Named Entity Recognition (NER) and Relation Extraction (RE) relied on hand-crafted rules, which were later replaced by probabilistic models such as Hidden Markov Models (HMMs) [8] and Conditional Random Fields (CRFs) [9]. Mikolov et al. [10] introduced distributed word representations, marking one of the first significant advances in capturing word similarity, which remains foundational in modern NLP. Lample et al. [11] proposed a BiLSTM-CRF architecture for NER, where a Bidirectional Long Short-Term Memory network is inserted between the input words and the CRF output layer.

Relation Extraction first appeared prominently in SemEval-2010 Task 8 [12], which focused on the “Relation Classification Subtask” and assigned a single label to a marked entity pair. The deep learning era began with convolutional neural networks (CNNs), which enabled mapping entire sentences to relation labels without manual feature engineering [13]. Subsequently, the advent of pre-trained language models (PLMs) marked a major advancement in RE, as highlighted by Li et al. [14]. These fine-tuned models, such as BERT and RoBERTa, trained on diverse datasets, have become standard for many modern NLP tasks.

3. Methodology

NER models are widely used in data mining, textual analysis, and text processing. However, they often lack flexibility, and training them can be a challenging task. To address these limitations, Zaratiana et al. [15] proposed GLiNER, a recent and effective alternative to traditional NER models, which are typically restricted to predefined entity types and rely on expensive large language models (LLMs). GLiNER is a NER model capable of recognizing a wide range of entity types using a Bidirectional Transformer architecture.

The model takes as input a set of entity type prompts and a sentence or text, with each entity type separated by a learned token [ENT] (see Figure 2). It uses a Bidirectional Language Model (BiLM) to generate token embeddings, which are passed through a feedforward network. Simultaneously, the input text is processed by a span representation layer. A matching score between the entity representations and the span representations is then computed using the dot product, followed by a sigmoid activation function [15].

In our implementation to perform NER on the GutBrain datasets, we used NuNER Zero, a zero-shot Named Entity Recognition (NER) model³. NuNER Zero is based on the GLiNER architecture and expects input as a concatenation of entity types and the target text. It was trained on the NuNER-v2.0 dataset⁴,

³https://huggingface.co/numind/NuNER_Zero

⁴<https://huggingface.co/numind/NuNER-v2.0>

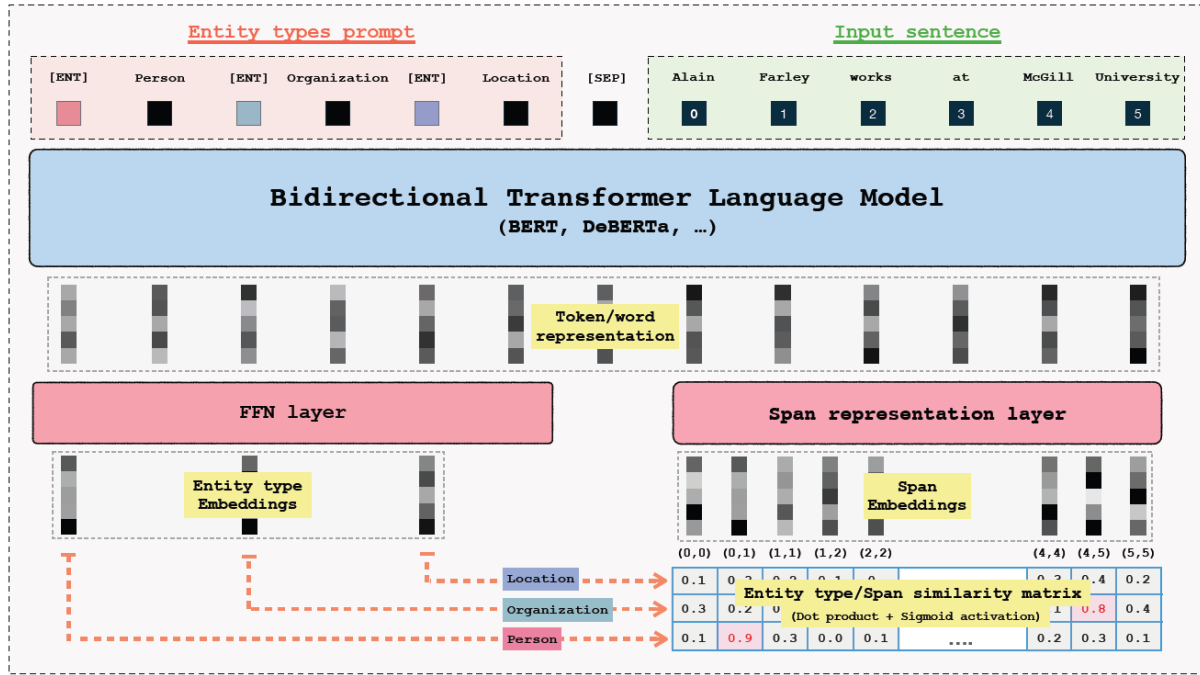


Figure 2: GLiNER Model Processing, from the GLiNER paper (<https://arxiv.org/abs/2311.08526>).

which combines annotated subsets of the Pile⁵ and C4⁶ corpora. Annotations were generated using large language models following the NuNER procedure [16].

Our implementation pipeline follows these steps:

- **Data analysis:** we begin by examining the composition of the GutBrain corpus (titles, abstracts, and entity annotations), removing any detected noise.
- **Data loading:** load the JSON files containing metadata and character-level entity labels.
- **Tokenizer initialization:** initialize the microsoft/deberta-v3-large tokenizer⁷, which splits the text into tokens and provides offset mappings for model input.
- **Preprocessing:** combine the title and abstract for each document; convert character-level entity spans into token-level spans using the tokenizer’s offset mappings; store the tokenized text and corresponding entities.
- **Hyperparameter configuration:** define the number of training steps based on dataset size, along with the maximum number of epochs, batch size, and maximum token length.
- **Model setup:** initialize the token classification model numind/NuNER_Zero based on GLiNER with pre-trained weights, and configure sampling parameters to manage training complexity.
- **Training loop:** iterate over batches, perform forward passes, compute loss, and apply backpropagation. Update model weights using the AdamW optimizer⁸.
- **Evaluation:** periodically evaluate model performance on dev . json, reporting Micro and Macro Precision, Recall, and F1 scores.

To predict NER entities with our trained models, we followed the following pipeline: for each article, we ran the model separately on its title and abstract, requesting predictions for all thirteen entity types (e.g., animal, anatomical location, DDF). The raw output consisted of a list of spans with start and end offsets, labels, and confidence scores. We then merged any adjacent spans with the same label and

⁵<https://huggingface.co/datasets/EleutherAI/pile>

⁶<https://huggingface.co/datasets/allenai/c4>

⁷<https://huggingface.co/microsoft/deberta-v3-large>

⁸<https://pytorch.org/docs/stable/generated/torch.optim.AdamW.html>

converted the character-level offsets into the required format. Finally, we saved the predictions in a JSON file, including the (start_idx) and (end_idx) fields.

For the second task of the GutBrainIE challenge, we employed two different pre-trained models: NeuML/pubmedbert-base-embeddings⁹, a fine-tuned model based on sentence-transformers trained on a dataset of randomly sampled PubMed¹⁰ title–abstract pairs and similar titles; and microsoft/BiomedNLP-BiomedBERT-base-uncased-abstract-fulltext¹¹, a biomedical model pre-trained from scratch using PubMed abstracts and full-text articles from PubMed Central¹² [17]. Our pipeline for this task is as follows:

- **Data loading:** we first load the JSON files containing the GutBrain datasets (platinum, gold, and development collections) and extract article texts, entity annotations, and relation annotations. We then initialize the microsoft/BiomedNLP-BiomedBERT-base-uncased-abstract-fulltext tokenizer to prepare the input for the neural network, inserting special entity markers—[E1] . . . [/E1] for subjects and [E2] . . . [/E2] for objects—into the raw text.
- **Preprocessing:** split the entities between the title and the abstract, and filter for relations involving entities from both.
- **Model configuration:** extend the BERT embedding layer to recognize the newly introduced entity markers. Instantiate the BiomedBERT model for sequence classification, where each input sequence (text with entity markers) is processed to determine the presence of a binary relation.
- **Training setup:** configure the AdamW optimizer and learning rate scheduler, and define the hyperparameters used to train the model, including number of epochs, batch size, learning rate, and sequence length.
- **Model training:** in each training step, perform a forward pass, compute the loss, backpropagate gradients, and update model weights via the optimizer and scheduler.
- **Validation:** after each epoch, evaluate model performance on the test split, reporting Micro and Macro Precision, Recall, and F1-score.

To predict relations between entities in the test dataset, we implemented the following pipeline. Starting from the named entities identified in Subtask 6.1, we generate all ordered pairs of distinct entities (e_1, e_2). We insert special markers ([E1], [E2]) into the raw text to highlight the subject and object, respectively. The marked text is then tokenized into a fixed-length sequence and converted into BERT input tensors. We feed the encoded sequence into a fine-tuned BertSequenceClassification model, whose output logits correspond to the relation labels defined in the label2id.json file. Finally, we apply the softmax function, select the top-scoring label with a confidence above 0.7, and save the predictions.

4. Experimental Setup

The experimental setup for this project includes the following components:

- The project source code is available in the ataudp2425-gainer repository on GitHub: <https://github.com/Vezzero/ataudp2425-gainer>.
- The dataset collection was provided by the CLEF BioASQ 2025 organizers and is available at: <https://hereditary.dei.unipd.it/challenges/gutbrainie/2025/>.
- The evaluation script used is evaluation.py, made available by the organizers via their official GitHub repository: https://github.com/MMartinelli-hub/GutBrainIE_2025_Baseline/blob/main/Eval/evaluate.py.

⁹<https://huggingface.co/NeuML/pubmedbert-base-embeddings>

¹⁰<https://pubmed.ncbi.nlm.nih.gov/>

¹¹https://huggingface.co/microsoft/BiomedCLIP-PubMedBERT_256-vit_base_patch16_224

¹²<https://pmc.ncbi.nlm.nih.gov/>

Table 5

Results of the 6.1 subtask.

Run ID	Training Dataset	Marco P.	Macro R.	Macro F1	Micro P.	Micro R.	Micro F1
ma	Platinum, Gold	0.580758	0.532186	0.528085	0.833333	0.739693	0.783726
md	Platinum, Gold, Dev	0.405397	0.541578	0.456943	0.639738	0.710590	0.673305
ms	Platinum, Gold, Silver	0.388916	0.550468	0.451079	0.633216	0.724333	0.675716

- Model training and prediction for both NER and RE tasks were performed using the following hardware:
 - Tesla T4 GPU on Google Colab: <https://colab.research.google.com/>
 - Dual NVIDIA T4 GPUs on Kaggle: <https://www.kaggle.com/>
 - 8× NVIDIA A40 GPUs on DEI (Dipartimento di Ingegneria dell’Informazione) cluster: <https://docs.dei.unipd.it/>

A README file is provided in the GitHub repository with complete reproducibility instructions.

5. Results

In this section, we report and describe the results obtained from the evaluation on the test data.

The runs submitted have been evaluated by the organizers on a held-out test set of 40 expert-annotated articles. These annotations are used as ground truth to compute the performance metrics.

5.1. Named Entity Recognition Results

For the NER task, we trained three variants of the numind/NuNER_Zero model: PironA, PironD, and PironS. All models share the same architecture but differ in the training dataset, number of epochs, and training steps:

- The first model, PironA, was trained on the platinum and gold datasets with 6,000 steps, a batch size of 2, and a maximum token length of 356.
- The second model, PironD, was trained on the platinum, gold, and dev datasets, with 8,000 steps, a batch size of 2, and a maximum token length of 600.
- The third model, PironS, was trained on the platinum, gold, and silver datasets. It used 12,000 steps, due to the larger dataset size—over 800 documents and 15,000 entities in total.

The NER test results are reported in Table 5. The PironA model with the **ma** run achieves the highest Micro-F1 score with high Micro-Precision and Micro-Recall. In particular, its Micro-Precision indicates that almost all entities are labeled correctly, resulting in the fewest overall errors. PironD with the **md** run obtains the highest Macro-Recall with lower Precision, suggesting it captures more true instances while generating more false positives. In contrast, PironS with the last run, **ms**, performs worst on both Micro and Macro scores, likely a consequence of the noisier annotations in the silver dataset.

5.2. Relation Extraction Results

For the Relation Extraction task, we trained four different models, finetuning the BiomedNLP-BiomedBERT-base-uncased-abstract-fulltext¹³ and NeuML/pubmedbert-base-embeddings¹⁴. BiomedBERT was fine-tuned under three different configurations, each varying only in training data and hyperparameters.

¹³<https://huggingface.co/microsoft/BiomedNLP-BiomedBERT-base-uncased-abstract-fulltext>

¹⁴<https://huggingface.co/NeuML/pubmedbert-base-embeddings>

Table 6

Performance metrics of the Binary Tag-Based Relation Extraction subtask.

Run ID	Training Dataset	Macro P.	Macro R.	Macro F1	Micro P.	Micro R.	Micro F1
ba1	Platinum, Gold	0.199798	0.398256	0.250001	0.41195	0.56710	0.477231
ba2	Platinum, Gold	0.193153	0.402862	0.245832	0.40361	0.58009	0.476021
ba	Platinum, Gold	0.238069	0.367984	0.269853	0.48092	0.54546	0.511156
bd1	Platinum, Gold, Dev	0.175111	0.540041	0.249663	0.32323	0.69264	0.440771
bd2	Platinum, Gold, Dev	0.169758	0.546031	0.244914	0.31262	0.69697	0.431635
bd	Platinum, Gold, Dev	0.215928	0.540124	0.290181	0.38517	0.69697	0.496148
bp1	Platinum, Gold, Dev	0.281089	0.365235	0.295995	0.50000	0.51948	0.509554
bp2	Platinum, Gold, Dev	0.251913	0.378696	0.285441	0.47727	0.54546	0.509091
bp	Platinum, Gold, Dev	0.317064	0.325444	0.296847	0.61497	0.49784	0.550239
bs1	Platinum, Gold, Silver	0.173007	0.542975	0.246921	0.31262	0.69697	0.431635
bs2	Platinum, Gold, Silver	0.166350	0.548190	0.240690	0.30112	0.70130	0.421326
bs	Platinum, Gold, Silver	0.213547	0.538597	0.287483	0.37915	0.69264	0.490046

- The first model, PironBinaryTG, was trained with platinum and gold datasets, 5 epochs, a batch size of 8, a learning rate of 2e-5, and a negative ratio of 0.3.
- The second model, PironBinaryTGD, was trained with platinum, gold, and dev collections. It differs from the previous one in the number of epochs, 8, the batch size, 10, and the learning rate of 1e-5.
- The PironBinaryTGS model was trained with the platinum, gold, and silver datasets. It was trained using the same number of epochs as the previous one, a batch size of 12, and a negative ratio of 0.2.
- Finally, the PironBinaryTGDNeuml model was trained from pubmedbert-base-embeddings with platinum, gold, and dev collections.

As explained in Section 3, we base our relation extraction models on the entity spans produced by our three NER variants (**ma**, **md**, and **ms**). For each test document, we first apply one of these variants to identify and save entities and then feed them into the following relation extraction model to generate the final relation labels.

5.2.1. Binary Tag-Based Relation Extraction Results

Table 6 reports both macro- and micro-averaged metrics for all submitted runs. The **bp** run, trained on the combined platinum, gold, and dev relation sets, with the BinaryTGDNeuml classifier and the NER predictions of the **ma** run, achieves the highest Macro F1 and Micro F1. We attribute this strong performance to its underlying BiomedBERT encoder: initially pre-trained on millions of PubMed abstracts and then fine-tuned in a sentence-transformers framework, it produces 768 dimensional embeddings that encode biomedical semantics. These more accurate representations help the model distinguish valid relations and invalid ones better and assign higher confidence scores to its predictions. The **bd** runs group, based on the NER prediction in the **md** run, shows a high Macro and Micro Recall but low Macro and Micro Precision, suggesting it is a Recall-oriented model. The **ba** group shows the opposite: higher Macro and Micro Precision but lower Macro and Micro Recall, suggesting it is a Precision-oriented model. The **bs** runs group had the lower Micro and Macro Precision (even if the Macro and Micro Recall are similar to the **bd** group). This shows again that the silver dataset’s noise negatively influences the Precision value.

In the scatter plot, Figure 3, each point represents a run, with Micro Precision on the x-axis and Micro Recall on the y-axis. The Recall-oriented **bd/bs** runs group cluster in the top-left with high Recall and low Precision. The Precision-oriented **ba** group is located in the mid-left. The balanced ones, **bp** group, appears toward the bottom-right, corresponding to their highest Micro F1.

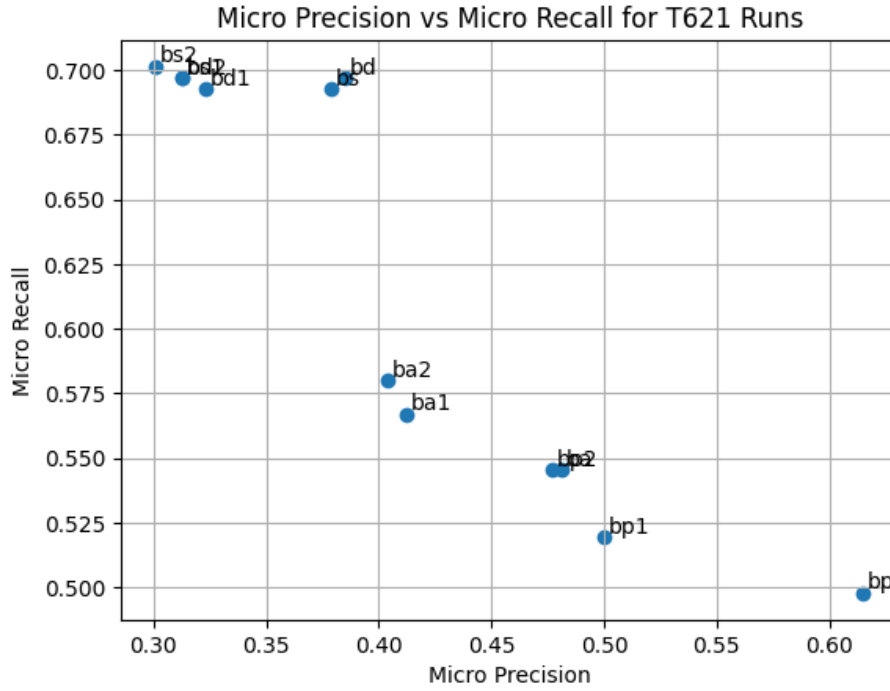


Figure 3: Scatter Plot of the Micro Precision and Recall of the 6.2.1 runs.

Table 7

Performance metrics of the Ternary Tag-Based Relation Extraction subtask.

Run ID	Training Dataset	Macro P.	Macro R.	Macro F1	Micro P.	Micro R.	Micro F1
ta1	Platinum, Gold	0.209531	0.178398	0.168514	0.581395	0.308642	0.403226
ta2	Platinum, Gold	0.209718	0.181461	0.168025	0.557971	0.316872	0.404199
ta	Platinum, Gold	0.254348	0.166327	0.181788	0.722772	0.300412	0.424419
td1	Platinum, Gold, Dev	0.286206	0.269013	0.259625	0.607362	0.407407	0.487685
td2	Platinum, Gold, Dev	0.271759	0.276622	0.252493	0.575419	0.423868	0.488152
td	Platinum, Gold, Dev	0.316739	0.231528	0.252816	0.740458	0.399177	0.518717
ts1	Platinum, Gold, Silver	0.271929	0.282908	0.258358	0.542373	0.395062	0.457143
ts2	Platinum, Gold, Silver	0.249736	0.286284	0.244771	0.500000	0.395062	0.441379
ts	Platinum, Gold, Silver	0.322955	0.257758	0.272830	0.681159	0.386831	0.493438

5.2.2. Ternary Tag-Based Relation Extraction Results

For subtask 6.2.2, we leveraged our PironBinary models, updating the inference pipeline to predict and return relations between entities. Table 7 reports the performance of each run.

The **td** run, obtained using the PironBinaryTGD model and the **ma** run, achieves the highest Micro Precision, resulting in the highest Micro F1. This suggests that the run correctly predicted nearly all relations between entities. In contrast, the **ts** run shows the best score in Macro Precision, as both rare and common relations contribute equally to the final average, leading to the best Macro F1. This comparison highlights that PironBinaryTGD provides the highest overall accuracy in the **td** configuration, while PironBinaryTGS, in the **ts** setup with fewer positive examples, excels in achieving balanced performance across all relation classes.

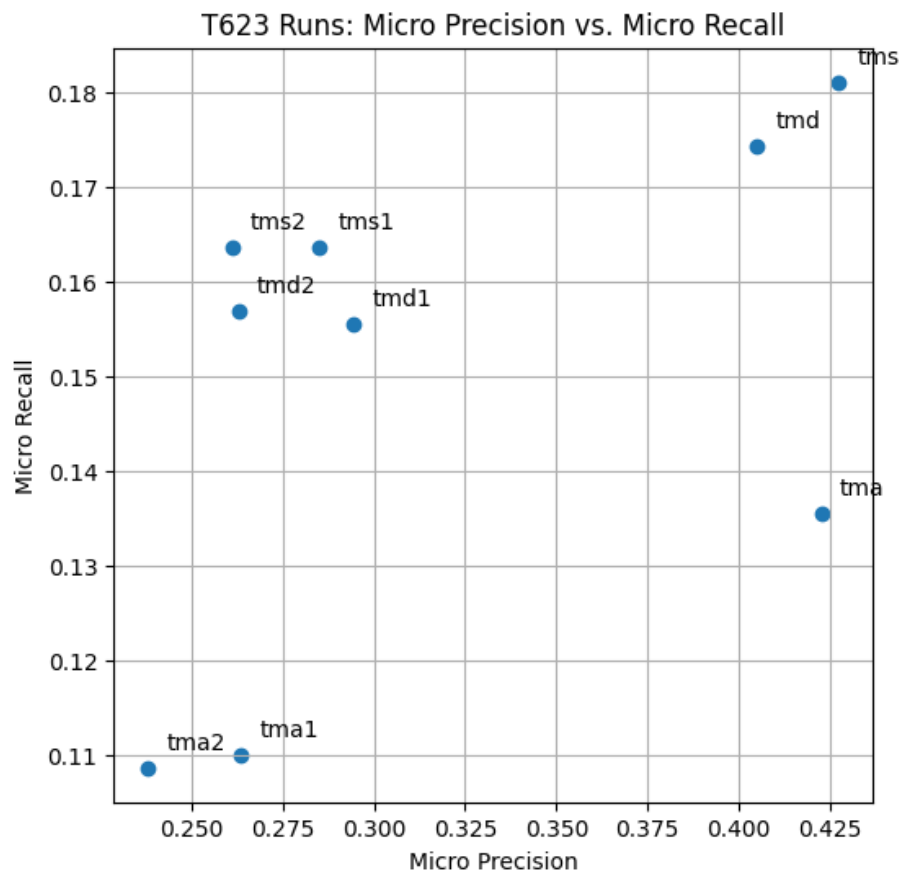
5.2.3. Ternary Mention Based Relation Extraction Results

Finally, Table 8 reports the performance of all 6.2.3 runs. The **ts** configuration, using the PironBinaryTGS model with the entities of the **ma** run, achieves the highest Micro and Macro F1, making it the best run

Table 8

Performance metrics of the Ternary Mention-Based Relation Extraction subtask.

Run ID	Training Dataset	Macro P.	Macro R.	Macro F1	Micro P.	Micro R.	Micro F1
tma1	Platinum, Gold	0.106581	0.063951	0.070624	0.263666	0.109920	0.155156
tma2	Platinum, Gold	0.095498	0.062951	0.065099	0.238235	0.108579	0.149171
tma	Platinum, Gold	0.165853	0.071890	0.090573	0.422594	0.135389	0.205076
tmd1	Platinum, Gold, Dev	0.146181	0.123795	0.127507	0.294416	0.155496	0.203509
tmd2	Platinum, Gold, Dev	0.149414	0.122917	0.117869	0.262921	0.156836	0.196474
tmd	Platinum, Gold, Dev	0.212696	0.125437	0.141929	0.404984	0.174263	0.243674
tms1	Platinum, Gold, Silver	0.159961	0.134892	0.132749	0.285047	0.163539	0.207836
tms2	Platinum, Gold, Silver	0.133641	0.134407	0.119694	0.261242	0.163539	0.201154
tms	Platinum, Gold, Silver	0.220283	0.138403	0.153766	0.427215	0.180965	0.254237

**Figure 4:** Scatter Plot of the Micro Precision and Recall of the 6.2.3 runs.

overall.

To explore how each run balances false positives and false negatives, Figure 4 plots Micro Precision against Micro Recall. In this scatter plot, the **tma** runs cluster in the upper left, with high Precision but low Recall, indicating they effectively avoid false positives but miss many true relations. The **tmd** group shows a well-balanced trade-off between Precision and Recall. The **tms** runs overlap with the **tmd** seeds, but the aggregated **tms** shifts into the top right quadrant, achieving the best combination of Micro Precision and Micro Recall—and consequently, the highest Micro F1.

6. Conclusions and Future Perspectives

In this paper, we presented the implementation pipelines for biomedical information extraction in the GutBrain corpus, covering both Named Entity Recognition (NER) and Relation Extraction (RE). For subtask 6.1, we fine-tuned the NuNer_Zero model under three hyperparameter configurations, achieving an almost 80% Micro F1 score. The chosen datasets used to train the model impacted the run performance, enabling us to see that the silver collections contained noisy annotations and leading to the worst results. In subtasks 6.2.1, 6.2.2, and 6.2.3, we carried out relation extraction as a marker-based sequence classification problem, using PironBinary models trained using the BiomedBERT base model. We were able to produce Precision-oriented, Recall-oriented, and balanced variants, each performing the best in different subtasks, but without a dominating configuration.

Looking ahead, we plan to investigate and test new LLMs' models and more noise-robust training techniques, such as confidence-aware loss functions, to improve both NER and RE performance.

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